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ORIGINAL ARTICLE

Efficient design of residential buildings geometry to optimize photovoltaic energy generation and energy demand in a warm Mediterranean climate

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Abstract The use of efficient design parameters in the initial stages of the life cycle of a building project helps to reduce the final energy demand. This article presents research results on the relation between the morphology of a building and its energy efficiency. Three types of residential buildings are analyzed: the single-family detached house, semidetached house and multidwelling building. The cases studied modeled in EnergyPlus to obtain building energy consumption per useful built surface. Also considered is the energy produced, thanks to the installation of photovoltaic solar panels on the building roof and on 50 % of the south façade surface. The paper provides a method to obtain the curve that shows the difference between the energy demand of residential buildings for various uses (HVAC, lighting, etc.) and the energy generated by installed solar panels in the building. The results reveal that the single-family detached housing model is the less energy-efficient. In the case of multidwelling houses, the optimal building height is obtained to reduce building energy consumption depending on total useful built area. The results show that up to 25 % of multidwelling building energy demand can be satisfied by solar energy on the rooftop and the façade. The balance between the energy demand and energy production of the building highlights the dimensional parameters that define optimal building shape from an energy efficiency perspective. The results obtained can be usefully applied to estimate the optimal geometric characteristics for a building of the same total surface area, which maximally reduces the final energy demand.

Keywords HVAC demand · Building shape · Building energy performance · Building energy simulation · Building-integrated photovoltaics (BIPV) · Building efficient design · Residential energy model

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Introduction

A major priority of the European Union (EU) is the implementation of strategies for smart, sustainable, and inclusive economic growth. In this context, the EU proposes the following objectives for 2020 (Commission of the European Communities March 2010):

- A 20 % reduction in EU greenhouse gas emissions from 1990 levels
- A 20 % energy consumption from renewable resources

- A 20 % reduction in the EU's energy consumption, by an improvement in energy efficiency of member countries (European Council September 2007)

The Europe's Renewable Energy Directive states that "energy efficiency and energy saving policies are some of the most effective methods by which Member States can increase the percentage share of energy from renewable sources" (European Commission 2009).

In recent years, European energy policies are diverting into two directions: on minimizing energy consumption and on reducing energy dependence and CO_2 emissions. This is the case of the Spanish Government actions, such as the Spanish Strategy for Energy Saving 2004–2012 and the National Action Plan for Renewable Energies (Spanish Strategy for Energy Saving 2004–2012) (Plan de Acción Nacional de Energías Renovables) (Institute for Energy Diversification and Saving 2007) (Institute for Energy Diversification and Saving 2010).

The use of solar photovoltaic energy is expected to increase dramatically all over the world in the coming years because of the rapid development of the photovoltaic industry and the continuous reduction of photovoltaic cost (Dinçer 2011). As an example, in the EU-27 the total amount of photovoltaic energy generated in 2010 (22.5 TWh) was 60.1 % greater than in 2009 (EurObserv'ER 2011).

In the case of Spain, residential buildings are responsible for up to 25 % of the national electricity demand (Institute for Energy Diversification and Saving 2011). An effective measure to reduce the energy demand in residential housing is the use of energy-efficient design criteria in the project design phase of building construction (Omer 2008; Tavares and Martins 2007). The optimization of the project from an energy-efficient perspective has been previously analyzed by various authors. There are numerous studies that estimate the energy demand of a building based on its geometric variables (compactness, shape factor, orientation, etc.) (Aksoy and Inalli 2006; Tzempelikos and Athienitis 2007; Florides et al. 2002; Lollini et al. 2010).

Depecker et al. (2001) obtained the relation between building shape and energy consumption from the values calculated by varying a shape coefficient defining the geometrical characteristics of the building. Ourghi et al. (2007) analyzed the impact of building shape on the total annual thermal needs of office buildings.

Ordóñez and Modi (2011) obtained the curve that represents CO_2 emissions for a building in which the gross overall surface is constant whereas the number of floors is variable. In this study, Ordóñez and Modi calculated carbon emissions by taking into account the energy consumption due to construction materials and the energy needs for heating, cooling and ventilation of residential building. As a result, for each case, the optimal number of building floors to reduce the energy consumption per built meter square was obtained. They also calculated the potential of CO_2 emissions reduction.

For building energy needs, solar energy is evidently one of the cleanest energy sources. In fact, according to Wang and Qiu (2009) it should be at the core of development programs. Similarly, after analyzing the residential photovoltaic (PV) market in France, Leloux et al. (2012) claimed that when optimally oriented, PV systems installed on buildings are a viable alternative to solar farms.

The energy balance between local PV panels and energy demand on residential buildings has been previously study by others authors. Carrilho da Graça et al. evaluated the technical and economic feasibility of solar Net Zero Energy Building (NZEB) systems in a single-family home. Two opposite houses are modeled: an efficient passive house and a glazed house, located in the mild southern European climate zone. Simulations are performed with EnergyPlus. Solar thermal panels and photovoltaic systems were sized to insure net zero energy demand of the models evaluated. After that, it is performed a cost/benefit analysis of each scenario, taking into account the government supported micro-generation payback schemes and including a 20-year net profit calculation. The researchers study deeply the thermal profiles and the demand loads, as well as the effect of building orientation on final energy demand (Carrilho da Graça et al. 2012).

Reynders et al. also evaluated the balance between domestic electricity use and local PV production, in this case for a detached single-family house in Belgium. However, in this study, the PV size is not constant, and it is performed for each individual case to achieve a predicted level of nearly zero energy building. They assessed the potential of using the thermal mass of a building as a structural thermal storage capacity to improve the cover factor of the PV system in the building. Six construction models with different levels of insulation are evaluated and the PV system is modeled in Modelica language. Finally, they evaluate the effectiveness of control strategies to reduce the grid dependence of the houses studied (Reynders et al. 2013).

The study described in the present paper analyzes the energy consumption of various geometries, which correspond to three conventional residential building types: detached house, semidetached house or townhouse, and multidwelling residential building. A curve is drawn for each building type, based on its energy needs.

Subsequently, a new variable is added, namely, the energy produced by photovoltaic solar panels on residential building roofs and on 50 % of the south façade surface. The energy balance is calculated as a function obtained as the difference between the energy consumption of heating and cooling systems and solar energy generation, based on the design parameters of the different building types. The set of values that minimizes this function defined the optimal building shape from an energy-efficient perspective.

The housing models in our study are defined on the basis of statistical data for the region of Andalusia (Spain). This information was obtained from the Spanish Ministry of Development (2010). As such, the profiles are regarded as representative of the residential sector. This makes it possible to compare real residential buildings with the building profiles in our study and to estimate the energy demand, depending on the building type. Figure 1 shows a flowchart of the procedure followed.

Description of dwellings

Building types, surface, distribution, and thermal zones

The housing models used in this study correspond to different archetypes. According to Swan and Ugursal (2009), this modeling technique represents the interconnectivity of appliances and end-uses within the house better than others. For this reason, it has been widely used to represent housing stock and determine building energy consumption. The archetypes can be defined by different criteria, such as the following: geometric characteristics, thermal characteristics, and operating parameters (Swan and Ugursal 2009).

The national report on building construction conducted by the Ministry of Development describes the residential sector in Spain, and provides information regarding the municipal construction permits issued by local authorities. The three residential building types focused on in the current study were the following:

- Single-family detached house
- Single-family semidetached house
- Multi-dwelling residential building

Table 1 lists the number of dwellings per building type as well as the average useful surface area per dwelling in

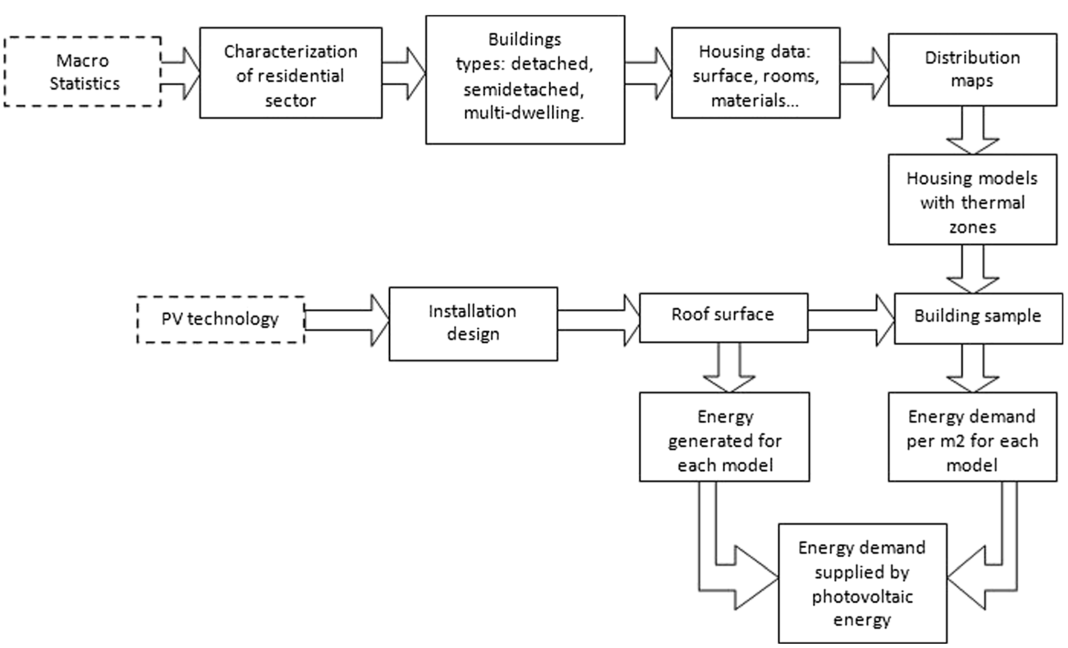

Fig. 1 Flowchart of the procedure

Table 1 Number of dwellings per building type, useful surface per dwelling and average values for each province in Andalusia (2007)

Province	No of dwellings per building			Useful surface area per dwelling (m^2)		
	Detached	Semidetach.	Multi-familiar	Detached	Semidetach.	Multi-familiar
Almeria	1,159	2,601	18,698	117.75	112.60	69.30
Cadiz	218	463	3,829	175.22	115.10	67.30
Córdoba	417	1,245	5,139	192.10	133.70	84.40
Granada	1,353	5,130	18,940	176.26	124.20	74.10
Huelva	168	833	5,241	143.98	135.80	77.10
Jaen	541	2,402	6,864	170.37	144.70	83.60
Malaga	1,590	2,259	20,094	152.40	125.10	70.40
Seville	1,391	3,241	7,780	132.37	113.90	72.30
AVERAGE (m^2)				152.54	124.47	73.29

each province and the average useful surface area per dwelling type.[1] The useful surface area per dwelling was distributed in functional units. In Andalusia, according to the statistical data analyzed, the average number of units is six rooms and three bathrooms in single-family detached houses, six rooms and two bathrooms in semidetached houses, and four rooms and two bathrooms in the case of multifamily residential buildings.

Based on these statistics, a ground-plan distribution model for each housing type was proposed. This distribution model gave the corresponding surface area for each functional unit. In order to reduce calculation time and following Ordenes et al. (2007), these functional units were classified in thermal areas. Figure 2 shows the distribution of each type of building and its corresponding division in thermal areas, as used in this study.

Collection of the sample and building properties

After the average useful surface for each residential housing type was calculated, the elements of the sample were obtained. For the single-family detached house, the sample consisted of only one element. Regarding the semidetached house profile, the sample was initially two houses and then increased to ten. Regarding the multifamily residential building, the following building profiles were defined: buildings with 5, 10, 15, 20, and 25 dwellings. The total useful surface area remained constant, whereas the number of floors and the height of the building were modified, so the building footprint also changes. An example is illustrated in Fig. 3 that shows a diagram of the building models corresponding to five multifamily dwellings in a residential building. The figure shows elevation plan, floor plant, and perspective view. Main dimensions and orientation are also indicated in the figure. Each element of the sample has a different compactness index, defined as the ratio between the volume and the exterior wall area of the building (Ourghi et al. 2007). As demonstrated in a previous study, the energy demand for heating and cooling increases with the number of building floors (Ordóñez and Modi 2011). Table 2 lists the geometric characteristics of the thermal areas for each residential building profile.

The surface and orientation of building openings have a bearing on building energy demand (Lollini et al. 2006; Manioglu and Yilmaz 2006). Albatici and Passerini (2011) found that there was a direct relation between south-oriented building façades and energy demand· Those elements of the sample with a rectangular ground plant (detached and semidetached houses) were oriented with their largest façade facing south, according to the guidelines for efficient design of buildings for heating dominated climate. In the case of multidwelling blocks, which had a square ground plant, all façades were of equal size, and they faced the cardinal points directly.

The *window to wall ratio* (WWR) is the ratio between the window surface and exterior wall surface of the building. For the multifamily buildings defined in this study, the

[1] Useful surface area should be determined as the gross internal floor area corresponding to the floor area contained within the building measured to the internal face of the external walls.

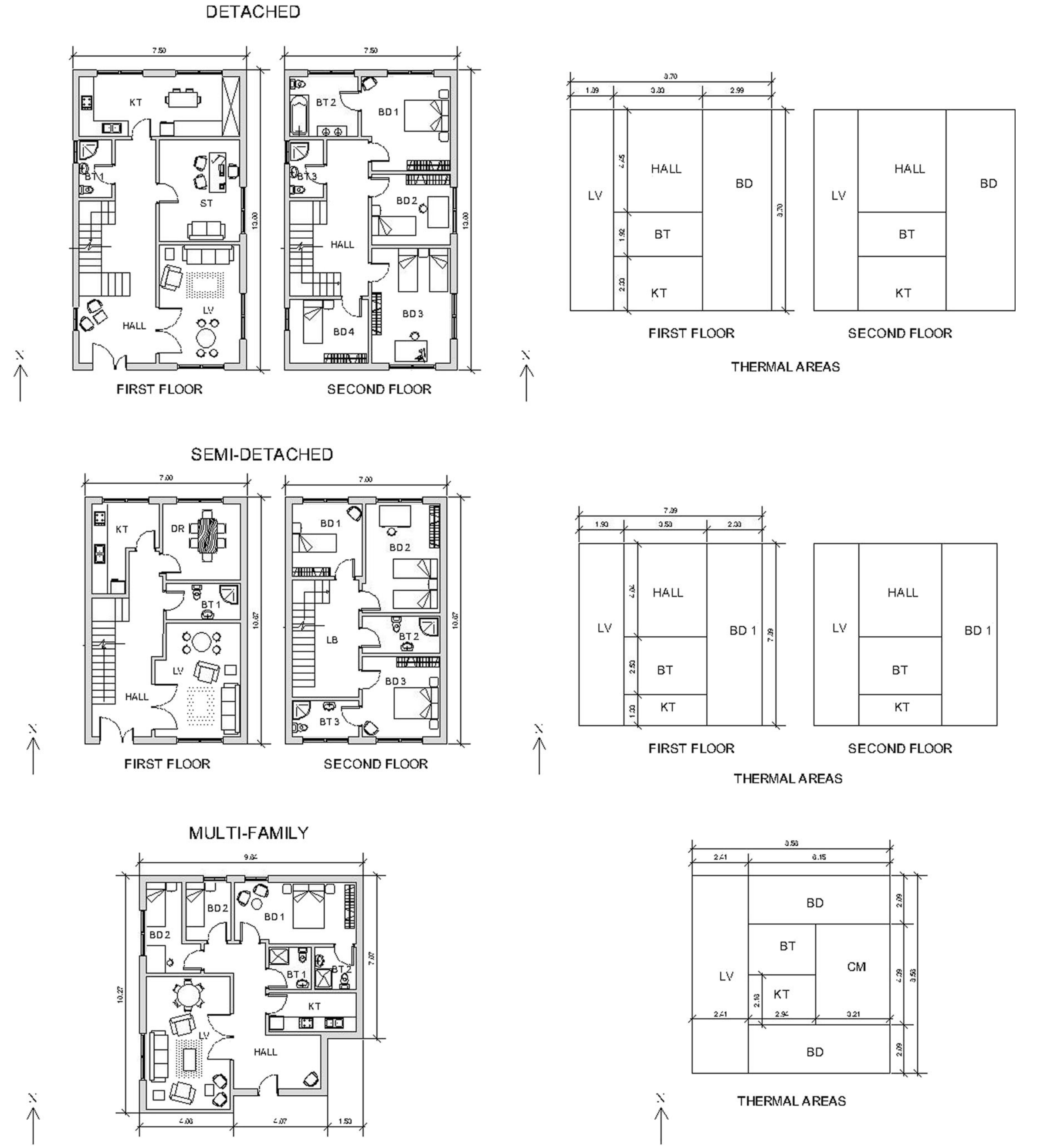

Fig. 2 Floor plans of each housing type and its equivalent in thermal areas. Main dimensions are indicated. Note 1: *LV* living room, *KT* kitchen, *BD* bedroom, *BT* bathroom, *CM* common spaces, *DR* dining room, *ST* study. Note 2: in detached and semidetached houses thermal models, the dimensions of thermal areas are the same in both floors

WWR remained constant and was the same as that obtained in the initial distribution. The window surface was then equally divided among the building façades.

Given that the objective of this research was to evaluate the relationship between the morphology of the building and its energy demand and the power-generating capacity of PV solar panels, no additional devices or passive design solutions were included. The finishing materials, construction systems, as well as the ventilation and infiltration

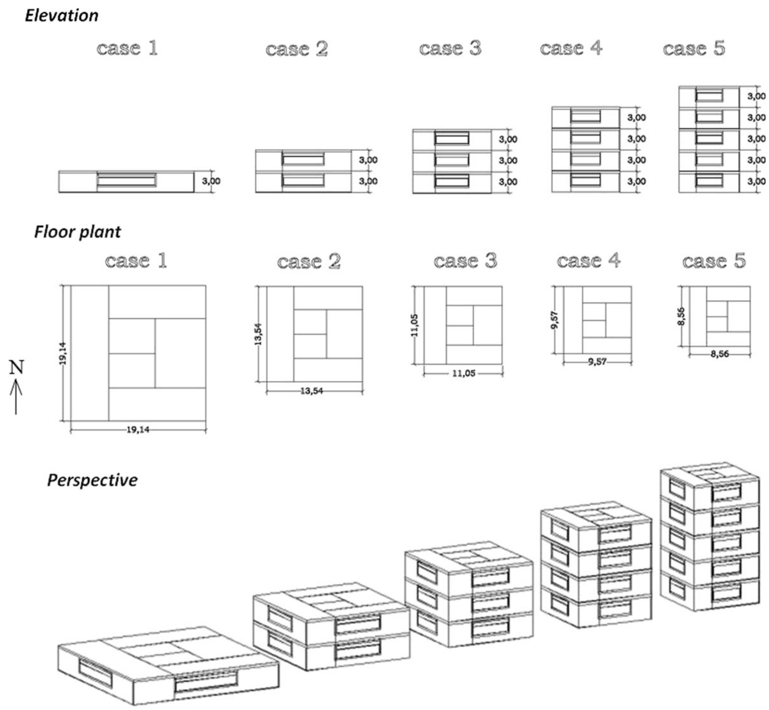

Fig. 3 Cases studies of a multifamily residential building corresponding to five dwellings. Case 1 is one floor height, case 2 is two floor heights, case 3 is three floor heights, case 4 is four floor heights and case 5 is five floor heights

rates applied in the simulation phase correspond to the most common values in Spain.

The total sample was made up of 85 building profiles: one profile corresponding to the detached building profile; nine to the semidetached profile; and 75 to the multifamily building profile. Tables 3 and 4 show the dimensional parameters of the profiles that made up the sample for each of the building types studied.

Table 2 Geometric characteristics of the thermal areas for each residential building profile

Building type	Thermal area	Area (m^2)	Volume (m^3)	Window to wall ratio (WWR) (%)
Detached	Living room	32.88	98.64	14.63
	Common area	34.00	102.00	
	Kitchen	17.86	53.58	
	Bedroom	51.98	155.94	
	Bathroom	14.78	44.34	
	Total	151.50	454.50	
Semidetached	Living room	15.22	45.66	25.00
	Common area	14.46	43.38	
	Kitchen	4.76	14.27	
	Bedroom	18.76	56.28	
	Bathroom	9.06	27.17	
	Total	62.26	186.76	
Multi-family	Living room	20.67	62.01	15.30
	Common area	14.67	44.01	
	Kitchen	6.36	19.08	
	Bedroom	25.07	75.21	
	Bathroom	6.53	19.59	
	Total	73.30	219.90	

Table 3 Dimensional parameters of the single-family detached house profile and the single-family semidetached house profiles

Case studied	No of dwellings	Total useful surface area (m^2)	Ground plan surface area (m^2)	Compactness (volume/envelope[a])	Façade/Floor (%)
Detached	1	151.50	75.75	1.26	137.88 %
Semidetached	2	249.00	124.50	1.40	114.07 %
	3	373.50	186.75	1.49	101.40 %
	4	498.00	249.00	1.54	95.06 %
	5	622.50	311.25	1.57	91.26 %
	6	747.00	373.50	1.59	88.72 %
	7	871.50	435.75	1.61	86.91 %
	8	996.00	498.00	1.62	85.55 %
	9	1 120.50	560.25	1.63	84.50 %
	10	1 245.00	622.50	1.64	83.65 %

[a] The building envelope includes surfaces exposed to external conditions: external walls, roof and basement

In order to elaborate models consistent with existing dwelling profiles, the most common finishing materials in building construction were selected, as reflected in the data. Table 5 shows the distribution of dwellings based on their structural solution and finishing materials, according to the statistics of the Ministry of Development in the region of Andalusia (Ministry of Development 2010). Also verified was compliance with national regulations regarding the limitation of the heat transfer coefficient U (W/m^{2}K) (Ministry of Development 2003a, b). Table 6 lists the thermophysical properties of the selected materials, including the calculated U values of the studied building constructions.

Methodology

Calculation of the energy demand

After specifying the building profiles and defining the construction material, it was necessary to calculate the energy demand per square meter of useful building surface. For this purpose, we used an energy analysis and thermal load simulation program. The use of building simulation software has increased in recent decades, especially in studies of energy efficiency in buildings.

In our study, the software utilized is EnergyPlus. EnergyPlus is a simulation tool for thermal analysis and energy load calculation, developed by the Department of Energy (DOE) of the US government and it combines the best features of previous programs (BLAST and DOE-2). This software application has been widely used in previous research studies to calculate the energy demand in buildings (Carlo and Lamberts 2008; Cook and Sproul 2011; Chan 2011; Cardinale et al. 2010; Lollini et al. 2010; Albatici 2009; Wang et al. 2009).

EnergyPlus uses hourly weather data to calculate thermal loads, and quantify the energy consumption due to heating and cooling, as well as the electrical system response. The integration of all aspects involved in the simulation allow designers to have a direct influence on a building's thermal response, instead of calculating loads and responses step by step (E.O.L. Berkeley National Laboratory 2010). More detailed information about EnergyPlus can be found in Crawley et al. (2001).

The models described in the section "Description of dwellings" (1 detached house; 9 semidetached houses, and 75 multifamily building buildings) were modeled in EnergyPlus. In all the cases, thermophysical properties of construction materials are the same (as previously explained) as well as the use of the building (residential). Water, lighting, and HVAC systems also have the same properties and regime of use in all the cases.

The geographic and meteorological data correspond to the city of Granada, characterized by a warm Mediterranean climate, according to the Köppen climate classification. The city of Granada is located at an altitude of 687 m.a.s.l and its exact geographic location is 37° 8′ 13″ N, 3° 37′ 53″ W. Monthly and annual mean climate values can be found in Table 7 (Spanish Meteorological Agency 2011).

Table 8 summarizes the values used for occupation rate, lighting installation power, source of artificial lighting and

Table 4 Dimensional parameters of multifamily building profiles with surface areas equivalent to 5, 10 15, 20, and 25 dwellings, depending on building height

Case studied	Building height (nº of floors)	Total useful surface area (m^2)	Ground plan surface (m^2)	Compactness (volume/envelope[a])	Façade/floor (%)
5-dwelling residential building	1	366.50	366.50	1.14	62.8
	2	366.50	183.25	1.59	88.65
	3	366.50	122.17	1.71	108.57
	4	366.50	91.63	1.71	125.36
	5	366.50	73.3	1.67	140.16
10-dwelling residential building	1	733.00	733.00	1.23	44.32
	2	733.00	366.50	1.84	62.68
	3	733.00	244.33	2.09	76.77
	4	733.00	183.25	2.16	88.65
	5	733.00	146.60	2.16	99.11
	6	733.00	122.17	2.11	108.57
	7	733.00	104.71	2.06	117.27
	8	733.00	91.63	2.00	125.36
	9	733.00	81.44	1.93	132.97
	10	733.00	73.30	1.87	140.16
15-dwelling residential building	1	1,099.50	1,099.50	1.27	36.19
	2	1,099.50	549.75	1.98	51.18
	3	1,099.50	366.50	2.32	62.68
	4	1,099.50	274.88	2.45	72.38
	5	1,099.50	219.90	2.48	80.92
	6	1,099.50	183.25	2.46	88.65
	7	1,099.50	157.07	2.41	95.75
	8	1,099.50	137.44	2.36	102.36
	9	1,099.50	122.17	2.29	108.57
	10	1,099.50	109.95	2.23	114.44
	11	1,099.50	99.95	2.17	120.03
	12	1,099.50	91.63	2.11	125.36
	13	1,099.50	84.58	2.06	130.48
	14	1,099.50	78.54	2.00	135.41
	15	1,099.50	73.30	1.95	140.16
20-dwelling residential building	1	1,466.50	1,466.50	1.30	31.34
	2	1,466.50	733.00	2.08	44.32
	3	1,466.50	488.67	2.48	54.28
	4	1,466.50	366.50	2.66	62.68
	5	1,466.50	293.20	2.73	70.08
	6	1,466.50	244.33	2.72	76.77
	7	1,466.50	209.43	2.69	82.92
	8	1,466.50	183.25	2.64	88.65
	9	1,466.50	162.89	2.58	94.02
	10	1,466.50	146.60	2.52	99.11
	11	1,466.50	133.27	2.46	103.95
	12	1,466.50	122.17	2.40	108.57
	13	1,466.50	112.77	2.34	113.00

Table 4 (continued)

Case studied	Building height (nº of floors)	Total useful surface area (m^2)	Ground plan surface (m^2)	Compactness (volume/envelope[a])	Façade/floor (%)
	14	1,466.50	104.71	2.28	117.27
	15	1,466.50	97.73	2.23	121.38
	16	1,466.50	91.63	2.18	125.36
	17	1,466.50	86.24	2.13	129.22
	18	1,466.50	81.44	2.08	132.97
	19	1,466.50	77.16	2.04	136.61
	20	1,466.50	73.30	2.00	140.16
25-dwelling residential building	1	1,832.50	1,832.52	1.32	28.03
	2	1,832.50	916.26	2.15	39.64
	3	1,832.50	610.84	2.60	48.55
	4	1,832.50	458.13	2.83	56.06
	5	1,832.50	366.50	2.92	62.68
	6	1,832.50	305.42	2.94	68.66
	7	1,832.50	261.79	2.92	74.17
	8	1,832.50	229.07	2.88	79.29
	9	1,832.50	203.61	2.82	84.10
	10	1,832.50	183.25	2.76	88.65
	11	1,832.50	166.59	2.70	92.97
	12	1,832.50	152.71	2.64	97.11
	13	1,832.50	140.96	2.58	101.07
	14	1,832.50	130.89	2.52	104.89
	15	1,832.50	122.17	2.46	108.57
	16	1,832.50	114.53	2.41	112.13
	17	1,832.50	107.80	2.36	115.58
	18	1,832.50	101.81	2.31	118.93
	19	1,832.50	96.45	2.26	122.19
	20	1,832.50	91.63	2.22	125.36
	21	1,832.50	87.26	2.17	128.46
	22	1,832.50	83.30	2.13	131.48
	23	1,832.50	79.67	2.10	134.44
	24	1,832.50	76.36	2.06	137.33
	25	1,832.50	73.30	2.02	140.16

[a] The building envelope includes surfaces exposed to external conditions: external walls, roof and basement

equipment installation power in each thermal zone. The dynamic procedure used by EnergyPlus takes into account heat gains from various elements, such as people, lighting, gas, and electrical appliances. Each of these internal sources is assigned a heat gain rate. The total heat gain is comprised of convective, radiant, and latent gains from these sources. The software applies usage schedules of occupancy, lighting, and use of appliances for each thermal zone. These schedules are used to estimate not only the heat flow generated inside the dwelling from the previously mentioned sources, but also the electricity consumption due to the usage of lighting systems and appliances.

As an example, Fig. 4 shows the occupation schedules and internal load patterns in kilojoules of two thermal zones, the living room and bedroom. It identifies people, lighting, and appliances as thermal sources. Differences in profile of room uses are appreciable. These internal gains influences on the energy balance and, consequently, on the

Table 5 Distribution of dwellings, based on their structure and finishes in Andalusia (2007)

Structure			
Vertical			
Reinforced concrete	Metallic	Load-bearing wall	Mixed and others
96.99 %	0.96 %	1.62 %	0.44 %
Horizontal			
Unidirectional		Others	
61.05 %		38.95 %	
Roof			
Flat		Pitched	
67.95 %		32.05 %	
Finishes			
Building envelope			
Ceramic	Stone	Siding	Other
55.63 %	6.87 %	35.21 %	2.30 %
Flooring			
Ceramic	Stone	Wood	Other
58.97 %	35.57 %	4.89 %	0.57 %

net heating, cooling and ventilation requirements, besides energy demand. The maximum and minimum HVAC set point temperatures used were 20 °C in winter and 24 °C in summer. Regarding the ventilation regime, general requirements of the Spanish legislation about minimum values have been met. According to the Spanish standard (Ministry of Development 2003b), the minimum vent flow for a living rooms is 3 l/s person, for a bed rooms is 5 l/s person, for kitchens is 2 l/s m^2 of useful surface, and in common zones is 2 l/s m^2 of useful surface. Natural ventilation is used in the buildings modeled, four air changes per hour. According to the Spanish Standard, the maximum air permeability of windows and doors value is 27 m^3/h m^2. The infiltration air flow rates obtained are 0.73 ach for detached houses, 1.82 ach for semidetached houses, and 1.46 ach for multifamily houses. As a reference, these results were compared with the average leakage levels in the Standard EN 15242 *Ventilation for buildings—Calculation methods for the determination of air flow rates in buildings including infiltration*. All cases studied are close to these average values and never exceed the maximum values.

When it was possible, standards values from the Spanish Legislation were adopted in decisions during the simulation process. If not, reference values were found in published literature on residential building simulation (Tavares and Martins 2007; Ordenes et al. 2007; Albatici 2009).

Calculation of the photovoltaic solar energy

Spain is one of the three countries in the EU where photovoltaic technologies have developed most rapidly, largely due to economic incentives and subsidies along with the development of a favorable regulatory framework (Ministry of Development 2003a, b; Prados 2010).

Andalusia is one of the regions in Europe with the highest potential for the production of solar energy (Ordóñez et al. 2010). In the first semester of 2011, Andalusia used renewable energies to produce a total of 4,376 MW of which 16.60 % (724 MW) corresponded to solar energy (Association of Promoters and Producers of Renewable Energies in Andalusia 2011).

In the current study, the energy generated has been estimated thanks to the installation of photovoltaic solar panels on the building roof and on 50 % of the south-oriented façade. A monocrystalline photovoltaic panel was selected for this purpose. After formulating the equation to calculate energy production, the tilt and orientation parameters were defined, as well as the percentage of solar radiation losses. Table 9 shows the properties of a monocrystalline photovoltaic panel under standard testing conditions.

The design method used to calculate the energy generated by the system is based on the following equation (Méndez and Cuervo 2008):

$$E_{\mathrm{p}} = G_{\mathrm{dm}}(\alpha, \beta) \times P_{\mathrm{mp}} \times \mathrm{PR}/G_{\mathrm{CEM}} \quad (1)$$

where E_{p} is the energy generated by the system in kilowatt hours per day; $G_{\mathrm{dm}}(\alpha, \beta)$ is the mean value of the daily irradiation on the solar generator in kilowatt hours per square meter per day; P_{mp} is the peak generator power in kilowatt hours under standard test conditions; PR is the performance ratio; and G_{CEM} is a conversion factor with value 1 kW/m^2.

The tilt angle β corresponds to the angle between the panel surface and the horizontal plane. Its value is between 0° and 90°. In this case, we selected the optimal tilt for an annual design period $\beta_{\mathrm{opt}}=\Phi-10$, where Φ is the latitude of the design location (Institute for Energy Diversification and Saving 2009). A lower tilt angle is not advisable because it favors the accumulation of dust on the panel (Leloux et al. 2012). The Azimuth angle α defines the angle between the projection

Table 6 Thermophysical properties of construction materials and systems used to define the model

Element	Thermal transmittance U (W/m^{2}K)	Layer	Thickness (cm)
Foundation slab	0.59	Ceramic flooring	3.0
		Cement mortar	2.0
		MW Mineral wool	5.0
		Reinforced concrete	50.0
		Mass concrete	5.0
		Sand and gravel	10.0
Intermediate floor slab	2.30	Ceramic flooring	3.0
		Cement mortar	2.0
		One-way concrete floor slab	30.0
		Plaster skimming	1.5
Traffic-bearing roof	0.37	Ceramic tiles	1.0
		Cement mortar	4.0
		Asphalt felt	1.0
		MW Mineral wool	8.0
		Asphalt felt	1.0
		Expanded clay concrete	10.0
		One-way concrete floor slab	30.0
		Plaster skimming	1.5
Outer double walls	0.68	Cement mortar	1.5
		Double-cavity brick	11.5
		Expanded polystyrene	3.0
		Air cavity	2.0
		Double-cavity brick	7.0
		Plaster skimming	1.5
Interior partition walls	2.60	Plaster skimming	1.5
		Double-cavity brick	7.0
		Plaster skimming	1.5
Window	3.43	Double glazing	0.6+0.6+0.4
		Metal framework	–

onto the perpendicular plane formed by the module surface and the meridian of the location, with values between −90 and 90°. A value of α=0 was used for the calculations.

The percentage of solar irradiation losses considered was 12 %. These losses can be due to module mismatch (3 %), soiling and nonperpendicular solar incidence (4 %) and shading (5 %).

The total number of solar modules was calculated, based on the roof surface of each of the building profiles. The distance d between panels satisfies the Eq. 2 (Ordóñez et al. 2010).

$$d = h/\tan(61^{\circ} - \Phi) \tag{2}$$

where h is the value for panel height with the optimal tilt and Φ is the latitude.

Analysis of variance test

An analysis of variance test (ANOVA test) was performed to test whether the output of the model (final energy demand) is independent from the building height or not. To do this, the 75 multifamiliar buildings of the sample were grouped into two classes depending on the

Table 7 Monthly/annual mean climate values in Granada, Spain

Month	Monthly/ Annual mean temperature (°C)	Monthly/Annual mean of the daily maximum temperatures (°C)	Monthly/Annual mean of the daily minimum temperatures (°C)	Monthly/ Annual mean precipitation (mm)	Mean relative humidity (%)	Monthly/ Annual mean number of frost days	Monthly/ Annual mean number of sun hours
Jan	6.8	12.2	1.3	44	74	13	161
Feb	8.4	14.1	2.6	36	69	6	161
Mar	10.7	17.0	4.3	37	62	2	207
May	12.6	18.8	6.4	40	59	0	215
Apr	16.5	23.1	9.8	30	55	0	268
Jun	21.3	28.8	13.9	16	48	0	314
Jul	25.3	33.5	17.1	3	41	0	348
Aug	25.1	33.2	17.1	3	42	0	320
Sept	21.2	28.5	14.0	17	52	0	243
Oct	15.7	21.9	9.5	40	64	0	203
Nov	10.6	16.2	5.1	46	73	3	164
Dec	7.9	13.1	2.8	49	76	8	147
Year	15.2	21.7	8.7	361	60	13	2,751

building height: low (from one to five floors) and tall (from 6 to 15). Figure 5 plots the least significant difference (LSD) intervals. The mean values of the two groups do not overlap and the p value is equal to 0. That means that there are statistically significant differences between the samples and that the final energy demand is influenced by building height.

Results and discussion

After running the simulations with EnergyPlus, the energy demand in kilowatt hour per square meter was obtained for each use (heating, cooling, lighting, etc.) and for each housing profile.

Cases 1 and 2: detached and semidetached single-family houses

Regarding the two types of single-family dwelling studied, the semidetached dwelling was found to be more energy-efficient than the detached dwelling. Despite the fact that the detached dwelling had a higher potential for the production of photovoltaic energy, the energy demand per useful surface was also greater.

In the most favorable scenario of semidetached dwelling (comprising ten items), a reduction in consumption of 26.67 % is obtained, compared to an isolated house. This is mainly because the detached house suffers higher heating losses because the proportion of façade is greater, and a considerable amount of heat losses occur through windows and external walls. As

Table 8 Occupation rate, lighting installation power, source of artificial lighting and equipment installation power

	Ocupancy	Lighting		Equipment
	Person/m^2	W/m^2	Source	W/m^2
Living room	0.30	20.0	Fluorescent lamps with ballast	5.0
Bedroom	0.16	4.4	Standard incandescent lamp	5.0
Kitchen	0.03	18.0	Fluorescent lamps with ballast	20.0
Bathroom	0.03	4.4	Standard incandescent lamp	3.0
Common zones	–	4.4	Standard incandescent lamp	–

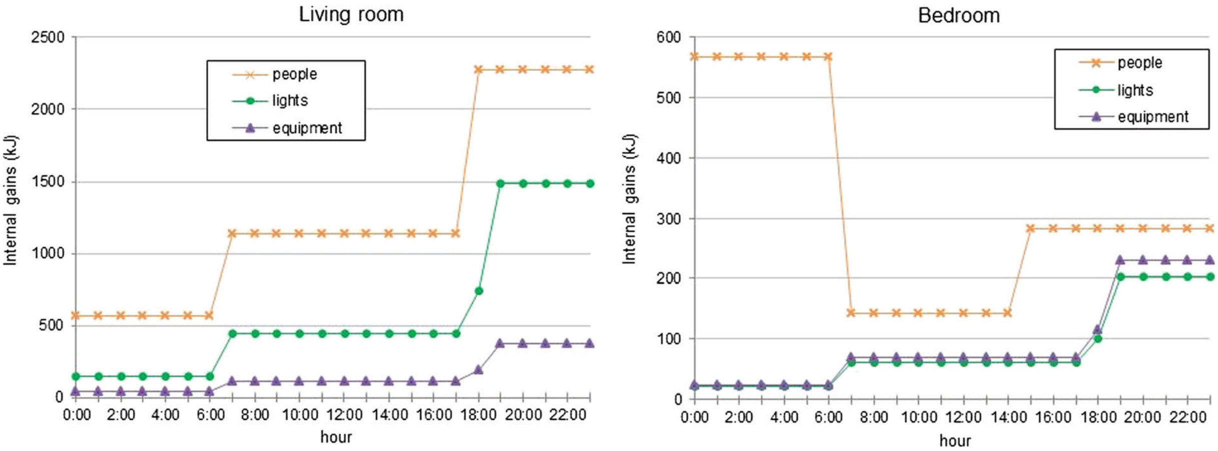

Fig. 4 Internal load patterns (kJ) for thermal areas due to occupation, lighting, and use of electrical equipment in two different thermal zones

can be seen in Table 3, the façade/floor rate in detached houses is higher than in all the cases of semidetached houses considered. In semidetached houses there are only two facades exposed to the outside, while the other two are internal walls (party walls) between heated spaces where heat losses are minimum. However, in the detached house four façades are exposed to the outside. This fact provokes a higher heating demand of the detached house per built surface. Heating losses can be reduced by improving the *U* values of external walls and windows. This improvement would reduce the differences in the energy demand of cases 1 and 2.

Table 10 shows the energy balance for cases 1 and 2. As can be seen, in detached houses the 21.64 % of energy demand is supplied by photovoltaic energy produced by the PV panels on the roof. In the case of semidetached houses, this share rises from 22.46 up to 26.67 % in the most favorable scenario.

Case 3: multidwelling residential building

Figure 6 shows on the left vertical axis the energy demand for multidwelling buildings (5, 10, 15, 20, and 25 dwellings) where the number of floors varies and the number of dwellings and useful surface area remain constant. The right vertical axis (in color green) shows is the energy produced thanks to the installation of photovoltaic solar panels on the building roof and on 50 % of the south-oriented façade.

As can be observed in Fig. 6 in the case of buildings with less useful surface area, the total energy demand increases with the height of the building. In contrast, in buildings with more useful surface area, such as those with 20 and 25 dwellings, the minimum energy demand corresponds to buildings with a height of two and three floors, respectively. As pointed out by Ordóñez and Modi (2011), when this optimal height is exceeded and the number of floors increases, so does the energy demand.

After calculating the energy balance between the initial energy demand and the energy generated in the building,

Table 9 Properties of the monocrystalline photovoltaic panel selected

Cell type	Dimensions (m)			Maximum power (W)	Open-circuit voltage (V)	NOCT[a] (°C)
	Width	Length	Thickness			
Monocrystalline with anti-reflexive layer (silicon nitride)	1.047	1.600	0.04	220±3 %	47.9	47

[a] *NOCT* Nominal Operating *Cell* Temperature, defined as the cell temperature under the conditions 800 W/m^2, 20 °C, AM 1.5, 1 m/s

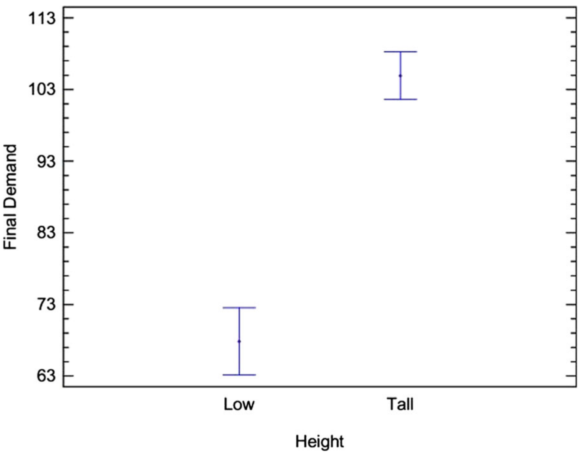

Fig. 5 LSD intervals (95.0 percentage) for the multifamiliar model

the final energy demand was obtained in kilowatt hour per square meter. It was then fit to a logarithmic function (see Fig. 7). The final consumption (i.e., the difference between the initial demand and the energy created by photovoltaic solar panels on the building) is given in (3):

$$\text{Final demand} = A{\cdot}ln(n) + B \tag{3}$$

where n is the number of floors; and A and B depend on the useful surface area of the building.

The graphs show that for the same height, the final energy demand varies, depending on the total useful surface area of the building. For example, given a five-floor building with five dwellings and a useful surface of 366.50 m^2, the estimated final energy demand is 112.86 kWh/m^2. In contrast, for a building of the same height but with 25 dwellings and a total useful surface area of 1,832.52 m^2, the estimated final energy demand is 79.23 kWh/m^2, almost 30 % less.

Figure 8 shows the percentage of the energy demand in multidwelling buildings (i.e., with 5 and 25 dwellings) which can be supplied by solar energy in relation to the building height. The blue bars represent the annual energy demand of the building per useful surface, and the red line represents the percentage of this demand that can be satisfied by the energy produced by photovoltaic panels.

The data show that, in the case of a multidwelling building with 25 dwellings and a total useful surface area of 1,832.52 m^2, up to 74 % of a building's energy demand can be satisfied with photovoltaic energy when the building has one floor. Needless to say, such a model is rarely found in the residential sector. Without going to such an extreme, percentages of roughly 25 % of the total energy demand supplied by photovoltaic energy can be reached in buildings of three floors, considered as a satisfactory result.

Modeling under different climatic conditions

To test the model potential, one of the building typologies has also been studied under different climatic conditions. Both the energy demand and the photovoltaic operational behavior are related to the climatic conditions and the solar irradiance level of the site. Cold and

Table 10 Energy demand (kWh/m^2), photovoltaic energy generated (kWh/m^2) and energy demand supplied (%) by photovoltaic system for detached and semidetached houses

Typology	Number of dwellings	Energy demand (kWh/m^2)	Photovoltaic energy generated (kWh/m^2)	% energy demand supplied by photovoltaic energy generated
Detached	1	130.08	31.44	21.64 %
Semidetached	2	110.13	29.74	22.46 %
	3	107.66	31.38	23.56 %
	4	106.42	33.15	25.03 %
	5	105.66	33.05	25.21 %
	6	105.16	33.95	25.93 %
	7	104.80	34.05	25.94 %
	8	104.53	34.60	26.39 %
	9	104.32	34.60	26.34 %
	10	104.15	34.79	26.67 %

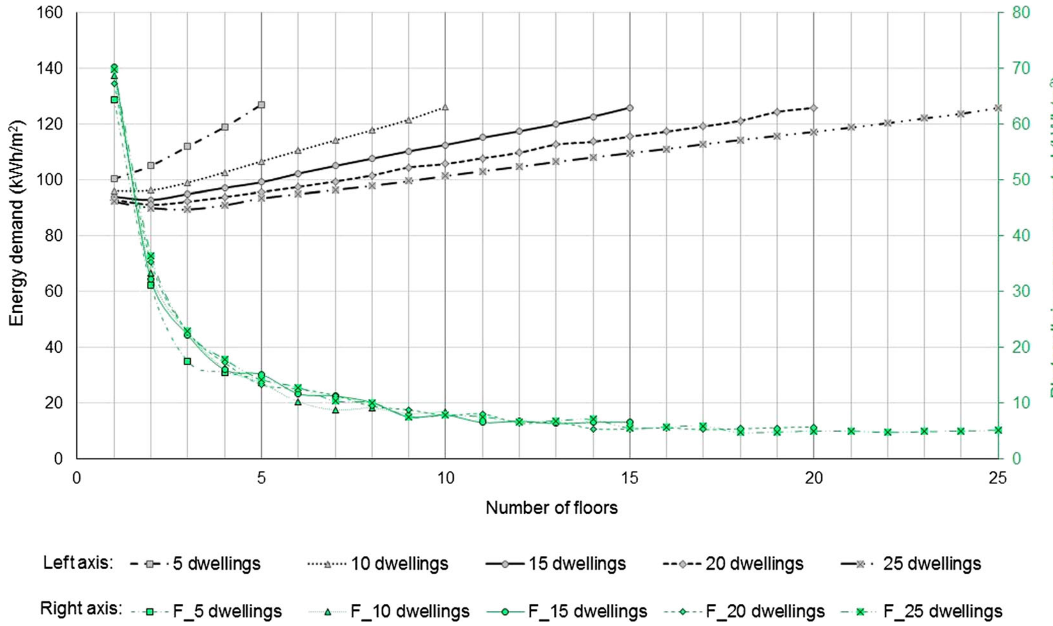

Fig. 6 Energy demand (*left vertical axis*) and photovoltaic energy generated (*right vertical axis*, *color green*) for each case of multifamiliar building studied

intermediate climatic conditions were selected to test the methodology and to compare with the results obtained for the city of Granada. The cold climatic site selected was the city of Oslo, in Norway. This site also represents a high latitude location. The intermediate city was Paris, warmer than Oslo but colder than Granada. The map of Fig. 9 shows the location of the cities considered and their global solar radiation levels. More details about the

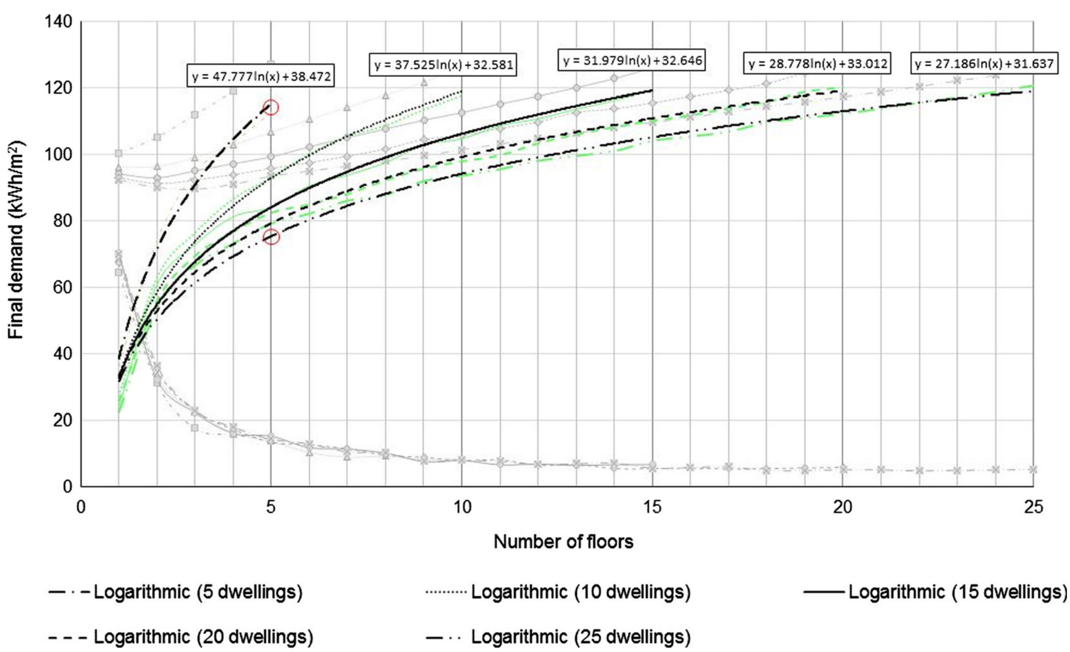

Fig. 7 Logarithmic equations that shows the difference between energy demand and energy generated, for each case studied

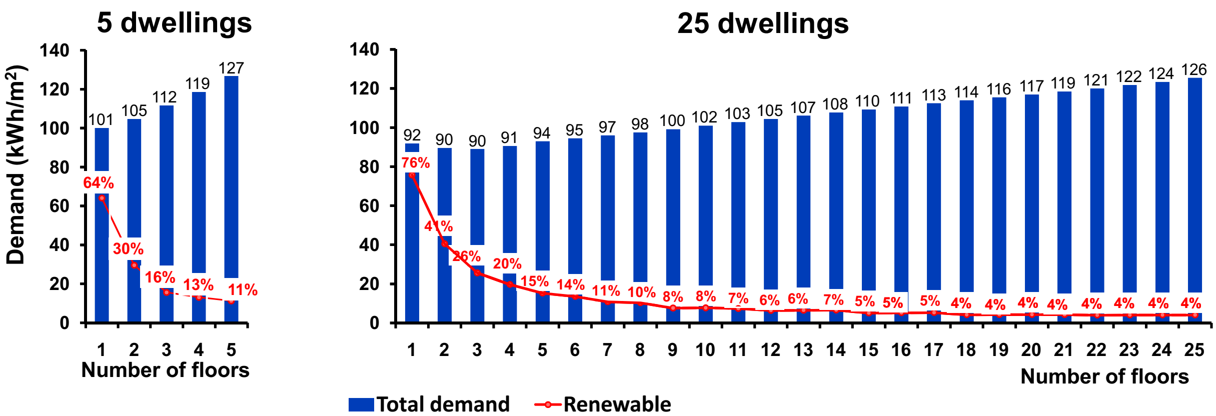

Fig. 8 Total energy demand of the building that can be supplied by solar energy, according to building height for a 5-dwelling residential building and 25-dwelling residential building

sites are given in Table 11 (Solar radiation database used: PVGIS-CMSAF http://re.jrc.ec.europa.eu/pvgis/apps4/pvest.php).

The prototype building was selected from the sample studied in the previous section and corresponds to a multifamiliar residential building consisting of 15 dwellings (1,099.50 m^2 useful surface area). This case was selected as it is an intermediate example of the multifamiliar residential sample studied in the section "Description of dwellings". Construction features, material

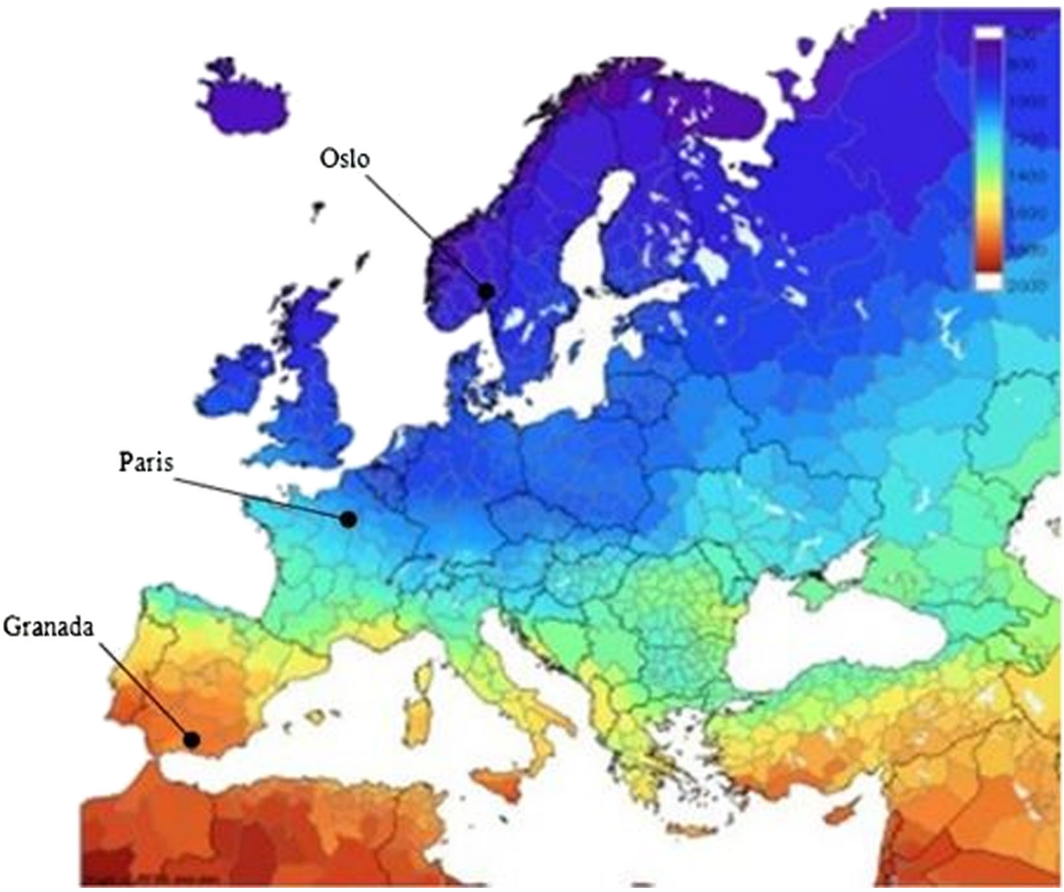

Fig. 9 Yearly sum of horizontal global irradiation (kWh/m^2/year) (source: http://re.jrc.ec.europa.eu/pvgis/solres/solreseurope.htm#Fig5)

Table 11 Climatic and solar radiation description of sites studied

Site	Latitude	Longitude	Elevation [m.a.s.l.]	I_{opt} [deg.]	Irr. Def. [%]	Köppen climate classification	Hh [Wh/m^2/day]	NDD
Granada	37° 8′ 13″ N	3° 37′ 53″ W	687	33	0.1	Csa: Hot-summer Mediterranean climate	5,160	1,318
Paris	48° 51′ 23″ N	2° 21′ 7″ E	41	36	0.2	Cfb: Temperate Oceanic	3,260	2,460
Oslo	59° 54′ 49″ N	10° 45′ 8″ E	20	40	0.1	Dfb: Warm summer humid continental climate	2,310	4,393

I_{opt} optimal inclination [deg.], *Irr. Def.* annual irradiation deficit due to shadowing (horizontal) [%], *Hh* irradiation on horizontal plane [Wh/m^2/day], *NDD* number of heating degree-days

properties, operational values and simulation approaches were the same as those explained in the previous sections.

Figure 10 represents the building specific power demand in kilowatt hour per square meter compared with the photovoltaic energy potential for each case study and each location considered. Results are compared with those obtained in Granada.

The simulation process reveals that, in consonance with Fig. 6, there is an optimal where the minimum total energy demand is reached. This optimal point corresponds to an optimal building height.

Regarding the photovoltaic energy generation, in a cold climate scenario (Oslo), the solar contribution to the energy balance is almost negligible due to the low level of solar radiation. The equation that represents the energy balance is exponential. However, in warmer climates, the contribution of solar energy is considerable, mostly in low buildings, and the balance equation is logarithmic.

An ANOVA test was performed to test whether climatic conditions have an effect on the final energy demand calculated by the model. Figure 11 represents the LSD intervals of the model outputs for the three cities studied. The statistical difference is confirmed. The intervals of mean values do not overlap, so the p value is equal to 0. This signifies that the population means are significantly different from each other and the model does not hold up well under different climate conditions. This conclusion is understandable because the results and the accuracy of the model would depend on the climatic zone for which it was designed and calibrated. The model considered was specifically designed for warm Mediterranean climates and locations with high solar radiation.

After obtaining these results, a third ANOVA test was also performed separately on each city, to evaluate if the results of the first ANOVA test performed on the sample in Granada (see Fig. 5) are consistent when other climatic conditions are considered. Figure 12 plots the LSD intervals of the model output with the buildings of the sample grouped into low and tall, both for Paris and Oslo. As in Granada, statistically significant differences are found and p value is equal to 0 in both cities. These results are consistent with findings in the first

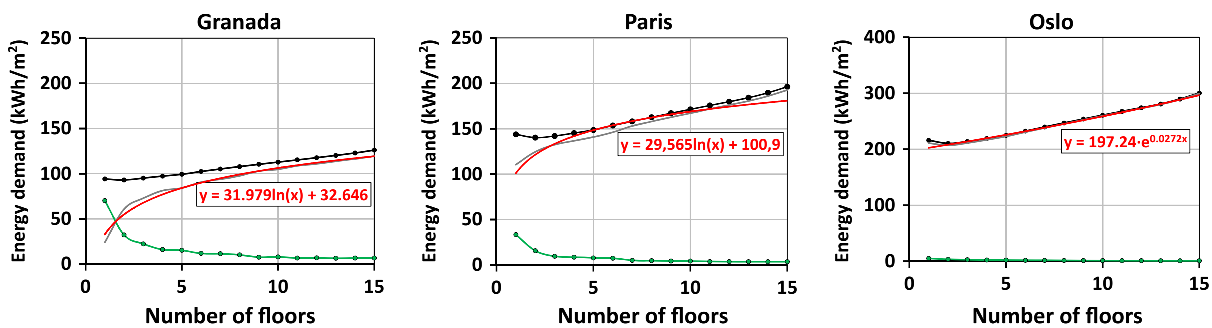

Fig. 10 Energy balance of the base case model in three cities studied. *Black line* is building energy demand, *green line* is photovoltaic energy generated, *gray line* is the energy balance and *red line* is the equation adjusted to the energy balance

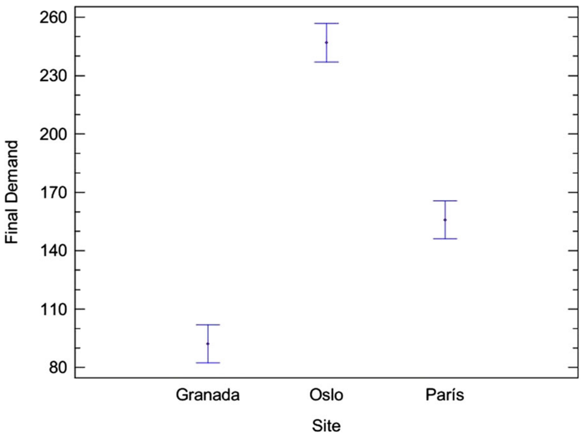

Fig. 11 Comparison of LSD intervals (95.0 percentage) for the model in the three cities studied

section and demonstrate that building height has an effect on the model results, independently of the climatic zone considered.

Conclusions

The authorities should promote the study and construction of energy-optimized housing, by integrating energy efficiency concepts in urban planning. It is generally accepted that the project design phase of building is the best stage to implement strategies to improve the energy performance of buildings throughout their lifespan. The effect of building shape on energy demand has been widely studied in terms of a shape coefficient that defines the geometric characteristics of a building.

Recent research studies have analyzed buildings from a broader perspective. Relevant variables include not only the energy demand but also other factors such as the environmental impact of the construction process. Thus, this study proposes a new variable, namely, the energy produced through the installation of photovoltaic solar panels on the roof of a building and on 50 % of the south façade.

Consequently, three types of residential buildings are analyzed. The main conclusion is that, given the same total useful surface area, the single-family detached housing model is less energy-efficient than the single-family semi-detached house. For the same useful surface area, the semidetached housing model saves up to 25 % more energy in comparison to the detached house.

In the case of multidwelling buildings, it is possible to cover part of the energy demand by installing energy panels on the building roof and on 50 % of the south-oriented façade, although for high buildings, the quantity of solar energy produced is very small in comparison to the demand. For the optimal case of a multidwelling three-floor building, the generation of photovoltaic solar energy on the rooftop and the façade can satisfy 25 % of the total energy demand.

The results shown in Fig. 6 reveal that for each built surface, there is an optimal building height where the energy demand reaches the minimum value. The ANOVA test conducted on the multifamiliar residential building model reveals that the building height has an effect on the building energy demand. In addition, this

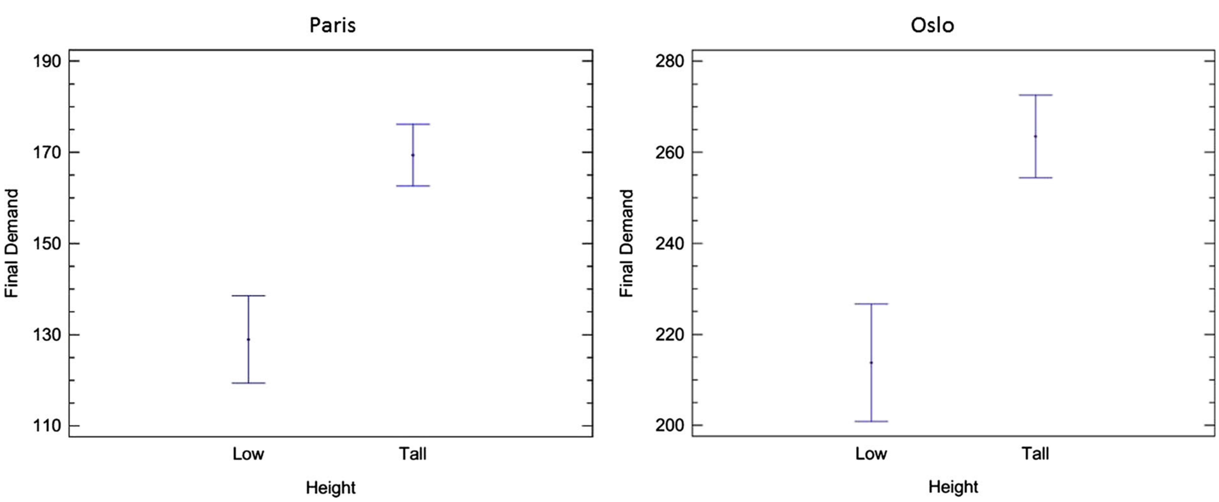

Fig. 12 LSD intervals (95.0 percentage) for the two height categories modeled in Paris (on the *left*) and in Oslo (on the *right*)

relation is also confirmed when other locations and climate conditions are considered.

This paper provides a method to obtain the curve that shows the difference between the energy demand of residential buildings for various uses (heating, cooling, lighting, etc.) and the energy generated by the installation of solar panels on the building.

The information thus obtained is useful for designers, engineers and other agents, not only during the conceptual design phase of the building but also in the planning of urban residential spaces. The results can be usefully applied to estimate the optimal geometric characteristics for a building of the same total surface area, reducing the final energy demand to a maximum.

From the design point of view, the impact on energy demand of the building height should be considered by designers. It is important to take into account the optimal building height during the design phase, to reduce the final energy demand of the building and to contribute to reducing carbon emissions. This purpose is well aligned with the aim of the European Union government as well as Environmental Associations, for the well-being of our planet.

Future research lines should study the impact of the insulation quality of materials, such as window glazing and frames, on the energy performance of the residential buildings studied. Also, innovative solutions that lower the window thermal conductivity should be considered, such as low emissivity glass coatings or window air gaps filled with argon.

Although these results of the research are satisfactory, the evaluation of the model in other cities and the ANOVA test conducted revealed that climatic conditions have an effect on the model results. Future research will analyze this effect and adapt the model to cold climates and low solar radiation locations.

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